

RAG based Language Agnostic Chatbot for Smart Query Response

Description:

Background: Campus offices answer hundreds of repetitive queries-fee deadlines, scholarship forms, timetable changes often from students more comfortable in Hindi or other regional languages. Long queues and communication gaps strain both staff and learners. **Description:** While answers reside in circulars and PDFs, students prefer conversational guidance. Generic LLMs struggle with hallucinated facts and lack access to internal, frequently updated institutional records. To solve this, a multilingual Retrieval-Augmented Generation (RAG) assistant that understands Hindi, English, and regional languages grounds responses directly in verified campus documents. Using cross-lingual dense vector embeddings, student inquiries in regional languages are accurately matched against English institutional notices. A stateful orchestration pipeline (built with LangGraph and LangChain) formulates factual responses citing specific circulars and dates. Key requirements include accurate intent recognition, context management across multiple turns, fallback to human contact when retrieval confidence is low, and daily conversation logs for continuous improvement. Embedding the chatbot on the college website and popular messaging platforms maximises reach. Crucially, the solution respects privacy, guarantees zero-hallucination accuracy through strict source grounding, and remains maintainable by student volunteers after the hackathon concludes. By focusing on automated content curation and user-centric design, participants can deliver high impact solution quickly.

Tech Stack:

- ReactJS (HTML CSS JS) for frontend
- Python for Backend (bs4, flask, langGraph, langChain)
- Vector Store & Multilingual Embeddings (ChromaDB / FAISS, bge-m3)
- SQL

Domain: Smart Education

Expected Outcome : Student innovators should develop a multilingual Retrieval-Augmented Generation (RAG) conversational assistant that ingests institutional circulars, PDFs, and FAQs into a vector index, recognises and responds to student queries in at least five regional languages, maintains context across follow-up questions, guarantees grounded responses with source citations, logs interactions for review, and integrates seamlessly into the college's existing web and messaging channels-achieving equitable, round-the-clock information access in a user friendly manner.